

Raphael Bernardo

Senior Android Engineer • *Ex-Uber (contract)*
Android TV, Fire OS and multi-brand OTT streaming — Media3/ExoPlayer, DRM, and native JNI/C++ (Uber Maps SDK).

LOC Rio de Janeiro • GMT-3
CIT Brazilian + Portuguese (EU) dual citizen
LANG Portuguese (native) • English (fluent) • Spanish (intermediate)
ENG Offshore B2B contractor (own CNPJ) • full US Eastern overlap
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§ 01 Summary

PROFILE • POSITIONING

Senior Android engineer with cross-stack reach — Uber Maps SDK (JNI/C++), OTT / CTV streaming frameworks, broadcast-grade DRM. Ten-plus years shipping production apps across Android, Android TV, Fire OS, webOS and Tizen — including a multi-brand OTT streaming framework and 4K DRM-protected playback (Widevine, PlayReady).

§ 02 Experience

ENGAGEMENTS • 2014 → PRESENT

Nexstar Media Group, Inc.

Aug 2025 – Present • current

Senior Fire OS / Android TV Engineer • via Nimble.LA • full remote **CONTRACT**

- Contributed to the architecture of an enterprise **OTT streaming framework** powering **multiple TV station brands** across Fire TV and Android TV using **MVVM + Clean Architecture** with a plugin-based integration layer for third-party services.
- Built five specialised **Media3 / ExoPlayer** implementations covering VOD, live, and audio playback with HLS adaptive bitrate.
- Shipped DRM-protected playback (**Widevine, PlayReady**) with both server-side and client-side ad insertion.
- Integrated the platform's analytics and audience-measurement stack under **GDPR / CCPA** compliance.
- Delivered TV-optimised auth (device-based), entitlement & flexible paywall, and zero-downtime feature rollouts across a large multi-brand flavor matrix via remote config; built Leanback TV UI with full d-pad navigation.
- Refactored callback chains to reactive Kotlin **SharedFlow**; resolved fragment-lifecycle leaks; wired **Firebase Crashlytics + New Relic** with JUnit, Mockk, Robolectric, Espresso coverage.

Kotlin Coroutines / Flow Media3 / ExoPlayer Dagger 2 Retrofit2 Moshi Glide
Android Leanback Jetpack (Nav, ViewModel, Room) Firebase Google IMA New Relic
JUnit • Mockk • Espresso

FreeCast, Inc.

Dec 2024 – Jul 2025 • 8 mo

Android / Android TV / Fire OS Engineer · full remote CONTRACT

- Returned to the Select TV and FreeCast Watch products across Android, Android TV, Fire OS, webOS and Tizen through to the end of the engagement.

Claro TV+

Apr 2024 – Dec 2024 · 9 mo

Android TV / Fire OS Engineer · full remote CONTRACT VIA GLOBAL HITSS

- Greenfield Android TV / Fire OS app for one of Brazil's largest pay-TV operators; **4K streaming** via DASH / HLS with multi-DRM.
- Contributed to the data layer as a shared **Kotlin Multiplatform (KMP)** submodule consumed by Android, webOS and Tizen clients.
- Multi-module **MVVM** architecture; Media3 playback; analytics across Firebase + Google Analytics 4.

Kotlin Multiplatform (KMP)

Kotlin

Jetpack Navigation · ViewModel · Room

Hilt

Media3

Coroutines / Flow

Firebase + GA4

webOS · Tizen

FreeCast, Inc.

May 2022 – Apr 2024 · 1 yr 11 mo

Android / Android TV / Fire OS Engineer · full remote CONTRACT

- Led the **full migration of the legacy Java codebase to Kotlin**; introduced Use-Case + Clean Architecture and improved end-user UX.
- Implemented **HLS / DASH** playback for **live linear TV and VOD** with Widevine DRM-protected streaming; integrated Google Cast and a paid subscription gateway.
- Maintained and developed companion apps on **webOS and Tizen** alongside the Android TV / Fire OS surfaces.
- Integrated **OTA tuner platforms** (HDHomeRun, AirTV, Alticast, PDAQ) with GStreamer-based media pipelines for secure live broadcast over HLS / DASH.
- Owned **DRM, EPG, tuner orchestration, transcoding**, and multi-device streaming end-to-end.
- Shipped on Google Play (*Select TV Android, FreeCast Watch*) and Amazon (*FreeCast Watch Fire OS*).

Kotlin

Jetpack Compose

Hilt

Coroutines / Flow

Media3

Room · DataStore

Retrofit2

Picasso

Firebase Analytics · Crashlytics · FCM

Mixpanel

webOS · Tizen

HDHomeRun · AirTV · Alticast · PDAQ

GStreamer

Widevine DRM

Uber Technologies

May 2021 – May 2022 · 1 yr

Android Engineer · Maps SDK team · via Matchpoint Solutions · full remote CONTRACT

- Worked on the **Maps team SDK** module — JNI/C++ bindings between Uber's native Maps engine and the Android client.
- Contributed to Uber's open-source Android architecture framework (**RIBs**); shipped to Uber Rider, Uber Eats and Uber Driver.
- Operated inside a 10K+ engineer codebase using **Buck, Flipper**, custom analytics / logger / crash reporter and a feature-by-module architecture.

Java · Kotlin

C++ · NDK · JNI

Groovy

Python · SQL

RIBs

Buck

CMake

Flipper

Firebase

FreeCast, Inc.

Mar 2021 – May 2021 · 3 mo

Android / Android TV / Fire OS Engineer · full remote CONTRACT

- Opening engagement on the Select TV Android and FreeCast Watch codebases, starting the migration from Java to Kotlin.

Super Revendedores

Oct 2018 – Feb 2021 · 2 yrs 5 mo

Android Engineer · full remote

- Migrated the full codebase from Java to Kotlin; implemented **online payment gateway**, virtual showcases and an SSO link to a partner web ordering portal.
- Built scalable asynchronous order processing on **AWS SQS / SNS** for reliable queuing and notification delivery.

Kotlin

AWS (SQS · SNS · Storage)

Firebase (Firestore · Auth · Remote Config · IAM · FCM)

Facebook OAuth

DbFlow

Dagger 2

MVP

Retrofit2

Concrete · an Accenture Company

Dec 2016 – Oct 2018 · 1 yr 11 mo

Jr. Android Engineer

- Built the **Minha CVC** travel app from its initial implementation as part of the agency team — **1M+ downloads**; shop geolocation directing customers to physical stores; itinerary feature reducing customer-service call volume.
- Worked on **Getnet**; migrated MVP to MVVM; introduced RxJava2 and LiveData.

Java 8

Kotlin

RxJava2 · LiveData

MVP → MVVM

Dagger 2

Google Maps API

Retrofit2 · Picasso

Firebase

Jenkins CI/CD

PraPagar (fintech)

Jun 2015 – Dec 2016 · 1 yr 7 mo

Jr. Android Engineer · collaborating with US team (MaxxPotential)

- Built mobile payments via **QR code and geolocation**; first Android + Laravel REST experience.

Android

PHP · Laravel 5

REST

OOP · Design Patterns

IBM Brasil

Jun 2014 – Dec 2014 · 7 mo

Software Engineering Intern · Rio de Janeiro · while pursuing B.S. at Instituto Infnet

- Wrote **SQL queries against IBM DB2** in support of contract and sales teams; produced reporting spreadsheets and Hyperion-based BI artefacts.
- Documented internal processes via SharePoint; first professional exposure to enterprise databases, REST APIs, and Agile workflows.

IBM DB2

SQL

IBM Hyperion

SharePoint

Python

REST APIs

Git

FASOLTI · Fábrica de Soluções em TI

Jan 2014 – May 2014 · 5 mo

Test Developer · INTERNSHIP

- Documented and implemented automated test cases in **Selenium with TestNG**.



Archive — Home Video Player

APPLE TV (TVOS)

Native SwiftUI tvOS client for a self-hosted media library: HLS playback, server-side transcoding, library browsing and resume across devices.

Swift, SwiftUI, tvOS, AVFoundation, HLS



Beer Price Calculator

ANDROID — GOOGLE PLAY

Compares beer prices by volume to find the best value per millilitre. Offline-first with local persistence and in-app billing.

Kotlin, Jetpack Compose, Room, Play Billing



Beer Price Calculator

IPHONE & IPAD (IOS)

Native SwiftUI build of the same price-per-volume comparison app for iPhone and iPad.

Swift, SwiftUI, iOS

Arquive — backend for the Archive tvOS app above

Self-hosted personal media archive & streaming server. In-browser **HLS playback** with GPU-accelerated transcoding cache, **IPTV** (M3U + XMLTV EPG + recording), DLNA/UPnP for smart TVs, distributed GPU-fleet transcoding, FAISS-backed face recognition, local-AI captions via Ollama, plus a native **Apple TV (SwiftUI)** client.

Python React HLS • GPU Transcoding
FAISS • InsightFace Ollama • Qwen2.5-VL DLNA
M3U • XMLTV SwiftUI tvOS

Socials Automator

End-to-end Instagram carousel generator — AI writes the captions, AI generates the images, then auto-posts. Live at [@ai.for.mortals](https://ai.for.mortals). Runs on **100% local AI** (LM Studio + ComfyUI) at zero cost, with cloud fallback (Z.AI, OpenAI, Groq, Gemini, fal.ai). Smart slide-count selection, post-history awareness, scheduled loop mode.

DynamicMusicApp

Android music player with **three pluggable design systems** users switch at runtime — *Material You* (Google), *Neon Wave* (synthwave), and *Organic Flow* (botanical). One Compose codebase, three radically different visual identities — design-system theming treated as a first-class architectural concern.

Kotlin Jetpack Compose Material 3 MVI
Coroutines / Flow Hilt

Infinite Research

Autonomous AI research system that refines documents through infinite self-learning iterations. **Agno agents** orchestrate **1–10 parallel DuckDuckGo searches** per iteration; each pass evolves search terms and patches identified gaps. Local LLMs via **LMStudio**, vector-DB RAG over prior refinements, SQLite versioning, graceful shutdown.

Python Agno Agents LMStudio (local LLM)
RAG • Vector DB DuckDuckGo Search SQLite

§ 05

Education

FORMAL · CONTINUING

Instituto Infnet · Rio de Janeiro

2013 – 2018

B.S. Computer Engineering — software development, mathematical modelling, OS & hardware architecture.

Chiswick House School (Malta)

1997 – 1998

Primary education (English-medium).

Orion Consulting · Rio de Janeiro

2010

Java fundamentals — OOP, software design, code patterns and logic.

§ 06

Technical Stack

REFERENCE · ATS INDEX

LANGUAGES	Kotlin Java C++ TypeScript JavaScript Python Groovy SQL PHP (Laravel)
MOBILE / TV	Android SDK Android TV Fire OS webOS Tizen Android Leanback Jetpack Compose Compose for TV Kotlin Multiplatform (KMP) NDK · JNI
STREAMING · MEDIA	OTT CTV (Connected TV) Linear TV VOD Live Streaming Media3 / ExoPlayer HLS DASH H.264 / AVC HEVC / H.265 Widevine PlayReady Google IMA SDK SSAI · CSAI GStreamer HDHomeRun · AirTV · Alticast · PDAQ
ARCHITECTURE	MVVM MVI Unidirectional Data Flow Clean Architecture Use Case Multi-module RIBs MVP Plugin-based integration
JETPACK · REACTIVE	Coroutines · Flow SharedFlow · StateFlow LiveData ViewModel · Navigation Room · DataStore Hilt · Dagger 2 RxJava2
CLOUD · BACKEND	Firebase (Analytics · Crashlytics · FCM · Auth · Firestore · Remote Config · IAM) AWS (SQS · SNS · Storage) REST APIs Retrofit2 · Moshi
ANALYTICS	Adobe Analytics Nielsen DCR Permutive BlueConic New Relic Mux Mixpanel Google Analytics 4
TESTING · BUILD	JUnit Mockk Robolectric Espresso Selenium · TestNG Buck Gradle CMake Jenkins CI/CD Flipper
PERFORMANCE · OBSERVABILITY	Android Profiler Firebase Crashlytics New Relic Playback lifecycle debugging Memory-leak analysis
COMPLIANCE · SECURITY	GDPR CCPA d-pad focus order DRM & Content Security OAuth · SSO